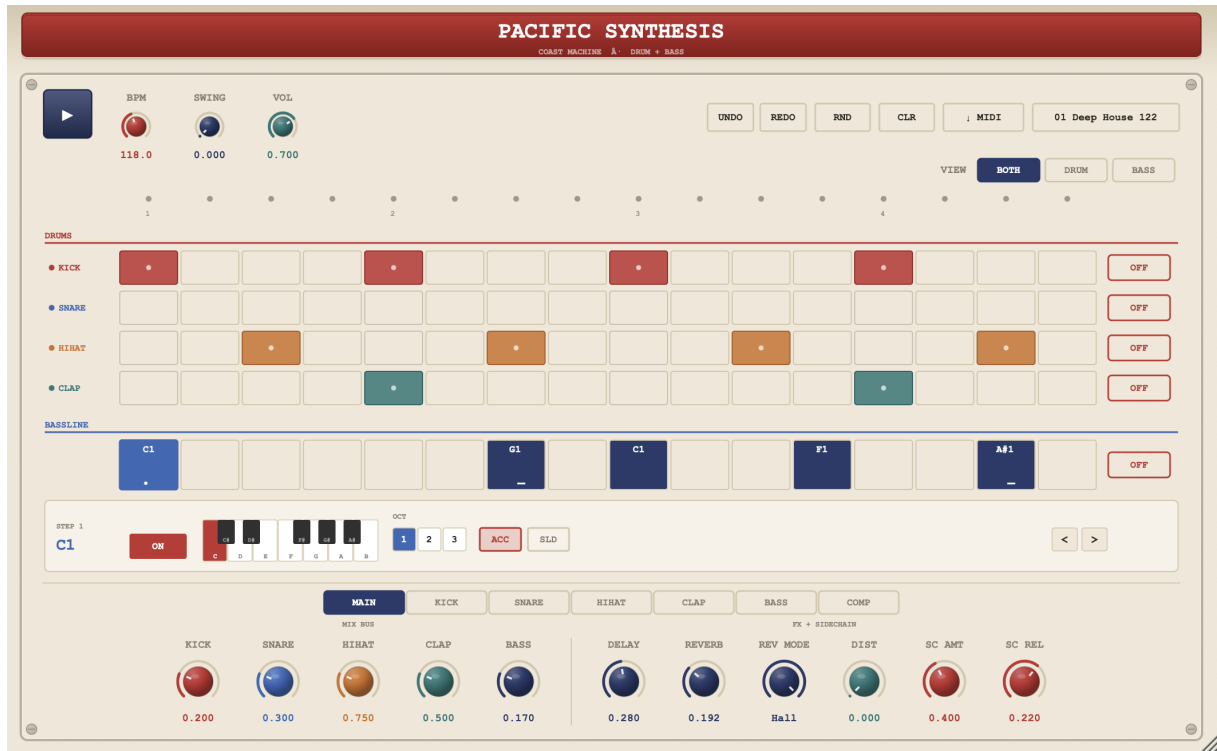


PACIFIC SYNTHESIS

User Guide — Sound Design Edition —



Coast Machine / Thee Coast Machine

v1.0 — 2026

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Introduction

Pacific Synthesis is a VST3/AU/Standalone plug-in that combines a step-sequenced drum machine and an acid-style bass synthesizer in one instrument. It models the classic TR-808/909/606 drum synthesis, a TB-303 style saw/square bass with Moog-style 4-pole ladder filter, and analog-flavored delay/reverb, making it well suited to Techno, House, Dub and Acid productions — without writing any code.

About This Guide

This PDF is the Sound Design Edition — a detailed parameter reference for all five voices: Kick / Snare / HiHat / Clap / Bass. For general operation, MIDI routing and host setup, please refer to the main user manual.

Each chapter is organized as follows:

- 1. Internal Structure — how the DSP works
- 2. Parameter Reference — every knob, in detail
- 3. Character Presets — 8 to 10 starting points
- 4. Tuning Order — what to move first when stuck
- 5. Troubleshooting — symptom to cause to fix

Color Coding

Each voice has its own accent color, matching the corresponding parts of the plug-in UI.

Color	Voice	UI counterpart
RED	Kick	Kick knob / pad
BLUE	Snare	Snare knob / pad
ORANGE	HiHat	HiHat knob / pad
TEAL	Clap	Clap knob / pad
PURPLE	Bass	Bass section

Chapter 1

Kick

1. Internal Structure

The Pacific Synthesis kick is a three-layer synth: sine oscillator + pitch envelope + click burst. Modelled after TR-808 / 909 analog kick synthesis, with fully tweakable modern controls.

Element	Role	Sonic character
Sine oscillator	Main tone (body)	"Boom" / "Thump"
Pitch envelope	Drops quickly from high to low — defines the punch	The "thoom" sweep
Click burst	Very short noise transient (~2 ms)	Initial "tick" attack
Drive (tanh saturation)	Adds saturation and harmonics at output	Thicker / dirtier weight

2. Parameter Reference

PITCH (30 — 140 Hz)

Final pitch (= bass tone) of the kick.

- 30 – 45 Hz: deep sub, almost felt (808-style, Dub, Trap)
- 45 – 60 Hz: standard Techno / House (909-leaning, Detroit)
- 60 – 80 Hz: punchy mid territory, less clashy with bass melody
- 80 – 140 Hz: higher and tom-like (Industrial / IDM)

Tip: The lower you go, the more it depends on the playback system. Verify on a small speaker.

AMOUNT (0 — 8 (octaves))

Pitch envelope depth — how many octaves above PITCH the oscillator starts before dropping.

- 0: no pitch motion, pure sine
- 1 – 3: subtle drop (Lo-Fi / Dub)
- 3 – 5: classic "thoom" (House / Techno)
- 5 – 7: aggressive punch (909 / Industrial)
- 7 – 8: extreme "zap"

Tip: PITCH and AMOUNT together define about 80% of the kick character.

DECAY (0.05 — 1.5 sec)

Sustain time of the sine envelope.

- 0.05 – 0.10: very tight (140+ BPM Hard Techno)
- 0.15 – 0.25: standard (House, mid-tempo Techno)
- 0.30 – 0.50: sustained, sub-like (Trap, Dub)

- 0.60+: long tail (808 melodic kick)

Tip: Long DECAY clashes with bass. Use sidechain (SC AMT) to duck the bass under the kick.

DRIVE (0 — 1)

Tanh soft saturation on the output.

- 0.0: clean
- 0.1 – 0.3: subtle harmonics, warmth
- 0.3 – 0.6: thick, forward (modern Techno staple)
- 0.6 – 0.9: obvious distortion (Industrial)
- 1.0: fully distorted

Tip: DRIVE emphasizes the pitch envelope. High AMOUNT + high DRIVE = "thwack".

3. Character Presets

Goal	PITCH	AMOUNT	DECAY	DRIVE
808 deep sub (Dub, Trap)	42 Hz	4.5	0.55	0.12
909 punchy (House, Techno)	60 Hz	4.8	0.40	0.35
Classic Detroit	50 Hz	4.0	0.30	0.25
606 tight (Electro)	75 Hz	3.5	0.18	0.30
Hard Techno (140+)	55 Hz	5.5	0.18	0.65
Industrial / distorted	50 Hz	6.0	0.25	0.85
Lo-Fi / Boom Bap	52 Hz	4.0	0.30	0.25
Dub Sub (bass replacement)	40 Hz	3.0	0.80	0.10
Acid Tech (Hardfloor)	46 Hz	3.8	0.24	0.25
Ambient (soft)	55 Hz	2.5	0.35	0.05

4. Tuning Order

#	Do this	Why
Step 1	PITCH (low-end weight)	Decide first how low the kick should sit
Step 2	DECAY (length)	Driven by tempo and surroundings
Step 3	AMOUNT (punch)	Punchy or mellow?
Step 4	DRIVE (character)	Forward or clean?
Step 5	VOL & PAN	Always judge against the full mix

5. Troubleshooting

Symptom	Cause & fix
Kick feels "not there"	PITCH too low. Check on a small speaker, or raise DRIVE for harmonics.
Kick floats / sits oddly	PITCH may clash with another instrument. Nudge ± 5 Hz.
Kick and bass mud together	Raise SC AMT (sidechain) to 0.3 – 0.5 to duck the bass.
Weak attack	Raise AMOUNT, or DRIVE above 0.3.
Too distorted	Lower DRIVE, or lower VOL (limiter may be reacting).
Loudness mismatch	Short DECAY = lower RMS. Use longer DECAY + DRIVE for level.

※ The kick is the mix reference. Lock it in first; placement of other voices becomes easier. Always tune in context, not in isolation.

Chapter 2

Snare

1. Internal Structure

A three-layer hybrid: noise + tonal oscillator + snap burst. Follows the TR-808 / 909 model, with the Crisp parameter taking you from full lo-fi to a sharp modern transient.

Element	Role	Sonic character
Noise (White / Pink)	Body source. NOISE selects the colour.	"shh" / "shhk"
Tonal oscillator	Fast pitch sweep 500 → 200 Hz over ~40 ms — provides body	"thock" / "doom"
Snap burst	Very short white-noise pulse right at the transient (1 – 5 ms)	"kah" / "pah"
1-pole HP filter (Crisp)	Highpass on the noise layer (30 Hz – 5 kHz)	lo-fi vs. crispy

2. Parameter Reference

DECAY (0.05 — 1.5 sec)

Total sustain time.

- 0.05 – 0.10: tight, tick-like
- 0.15 – 0.20: standard snare
- 0.25 – 0.40: full, sustained
- 0.50+: barely a snare anymore

Tip: Driven by tempo and space.

NOISE (0 = White, 1 = Pink)

Noise colour.

- White (0): flat / bright → 909 / Modern
- Pink (1): tilted low / warm → 808 / Vintage

Tip: Modern: white. Warm: pink.

TONE (0 — 1)

Mix amount of the tonal sine oscillator.

- 0.0: pure noise (clap-like)
- 0.3 – 0.5: balanced body
- 0.7+: tom-like
- 1.0: basically a tom

Tip: 0.4 – 0.5 is standard.

SNAP (0 — 1)

Very short noise burst at the trigger.

- 0.0: smooth attack
- 0.3 – 0.5: natural emphasis
- 0.8 – 1.0: sharp, in your face

Tip: Raise to cut through a busy mix.

CRISP (0 = Lo-Fi, 1 = Crisp)

1-pole highpass filter on noise.

- 0.0: 30 Hz cut (bypass)
- 0.3: ~450 Hz
- 0.5: ~1250 Hz
- 0.7: ~2500 Hz
- 1.0: 5 kHz cut

Tip: Lo-Fi: stay low. Modern: 0.5+.

3. Character Presets

Goal	DEC	NOISE	TONE	SNAP	CRISP
TR-808 (warm)	0.15	0.65	0.40	0.45	0.20
TR-909 (House / Techno)	0.18	0.35	0.35	0.65	0.45
TR-606 (tight)	0.08	0.50	0.50	0.55	0.60
Acid (sharp)	0.12	0.40	0.45	0.70	0.55
Industrial	0.15	0.30	0.55	0.75	0.65
Dub (soft, long)	0.30	0.70	0.40	0.20	0.05
Lo-Fi / Boom Bap	0.10	0.40	0.25	0.35	0.00
DnB / Jungle	0.10	0.40	0.45	0.75	0.65
Modern Tech House	0.13	0.45	0.45	0.60	0.55
Ambient	0.25	0.70	0.30	0.15	0.05

4. Tuning Order

#	Do this	Why
Step 1	Set DECAF	How long should it ring?
Step 2	Balance TONE	Want body, or not?
Step 3	Adjust brightness with NOISE	White vs. Pink
Step 4	Cut-through with CRISP	Muffled, or up front?
Step 5	Finish with SNAP	Fine-tune attack feel

5. Troubleshooting

Symptom	Cause & fix
Thin / weak snare	Raise TONE (0.4 – 0.6) and SNAP.
Clashes with the kick	Raise CRISP to highpass the noise away from the kick.
Too much "shh"	Move NOISE toward pink, lower CRISP, or shorten DECAF.
Sounds like a tom	Lower TONE below 0.5.
Weak attack	Push SNAP to 0.5+, raise CRISP.
Stands out on slow tracks	Pull SNAP and CRISP back, extend DECAF.
Distortion on rolls	Lower SnareVol or shorten DECAF.

※ The snare carries the backbeat. Together with the kick it defines the groove. When layered with the Clap, separate by PAN or FREQ.

Chapter 3

HiHat

1. Internal Structure

White noise + metal oscillators → bandpass → drive. Includes the TR-808 set of six square waves at 215/304/364/467/542/681 Hz.

Element	Role	Sonic character
White noise	Broadband base material	Soft hiss
Metal oscillator (6 squares)	TR-808 metallic resonances	"shing", "tk"
Bandpass filter	Picks out a frequency band (FREQ + Q)	Tonal character
Drive (tanh)	Saturation for aggression	Edgy, cutting
Closed / Open	MIDI notes select decay time (mutually exclusive)	Short vs. long

2. Parameter Reference

FREQ (3000 — 14000 Hz)

Center frequency of the bandpass.

- 3000 – 5000 Hz: darker, warmer
- 5000 – 8000 Hz: mid-tilted, felt-like
- 8000 – 10000 Hz: standard House / Techno
- 10000 – 14000 Hz: shimmering, Modern / EDM

Tip: Keep 2 – 3 kHz separation from Snare / Clap.

Q (1 — 12)

Bandpass sharpness.

- 1 – 2: broad
- 2 – 4: natural hat
- 4 – 7: pinpoint
- 7 – 12: heavy resonance

Tip: High Q reduces level. Compensate by lowering FREQ.

METAL (0 — 1)

Mix amount of metal oscillator.

- 0.0: pure noise
- 0.3 – 0.5: subtle edge (808-style)

- 0.5 – 0.8: clearly metallic
- 0.9 – 1.0: mostly metal

Tip: Pairs well with Q for character variation.

DRIVE (0 — 1)

Tanh saturation after bandpass.

- 0.0: clean
- 0.1 – 0.3: subtle weight
- 0.4 – 0.7: distinct distortion
- 0.8 – 1.0: noisy, rough

Tip: Can stand in for drum-bus tape saturation.

CLOSED DECAY (0.01 — 0.4 sec)

Closed hat decay.

- 0.01 – 0.03: very tight
- 0.03 – 0.06: standard closed
- 0.06 – 0.15: shuffle feel
- 0.15+: practically open

OPEN DECAY (0.05 — 1.5 sec)

Open hat decay. Triggered by MIDI note 92 (G#5).

- 0.05 – 0.15: short open
- 0.15 – 0.4: standard open
- 0.4 – 0.8: long, jazzy
- 0.8 – 1.5: very long tail

Tip: Closed/Open are mutually exclusive (closed chokes the ringing open).

3. Character Presets

Goal	FREQ	Q	METAL	DRIVE	C-DEC
TR-808 style (warm)	8500	3.0	0.30	0.00	0.04
TR-909 style	9500	2.5	0.45	0.10	0.045
TR-606 style (Electro)	7500	4.5	0.75	0.05	0.035
Acid / Hardfloor	8500	3.5	0.40	0.25	0.045
Industrial	8000	6.0	0.60	0.70	0.035
Footwork / DnB	10500	2.5	0.30	0.00	0.025
Lo-Fi / Boom Bap	6500	2.5	0.10	0.00	0.055
Modern Tech House	10000	2.2	0.35	0.10	0.045
Ambient / Dub	6000	2.0	0.10	0.00	0.080

4. Tuning Order

#	Do this	Why
Step 1	Place FREQ away from other top-end	Sit above Snare or Clap?
Step 2	Set CLOSED DECAY	Tight or shuffly?
Step 3	Dial METAL	Soft, or mechanical?
Step 4	Sharpen with Q	Broad or pinpoint?
Step 5	DRIVE for aggression	More cut, more grit

5. Troubleshooting

Symptom	Cause & fix
Hat gets buried	Separate FREQ by 2 – 3 kHz, or use DRIVE.
Painfully shiny	Lower FREQ, Q, or HhVol.
Too mechanical	METAL below 0.3, lower Q.
No open hat	Send MIDI note 92 (G#5) on ch 10.
Closed and open ring together	Mutually exclusive — intended choke behaviour.

※ The hi-hat carries the groove. If the kick is "boom", the hi-hat writes the "tick-tack".

Chapter 4

Clap

1. Internal Structure

Pink noise → bandpass + Style burst train. Switches between Single, TR-707 (×2) and TR-909 (×3) styles.

Element	Role	Sonic character
Pink noise	Soft base material (low-tilted noise)	Soft "pa"
Bandpass filter	Shaped by FREQ + Q	Tone of the instrument
Style burst	Switches Single / 707 (×2) / 909 (×3)	"papa" / "papapa"
Decay envelope	Determines the ringing time	Tight vs. rich

2. Parameter Reference

FREQ (500 — 3000 Hz)

Center frequency of bandpass.

- 500 – 800 Hz: dark (Dub, Lo-Fi)
- 800 – 1200 Hz: standard 808
- 1200 – 1800 Hz: bright (TR-909, House)
- 1800 – 3000 Hz: sharp (EDM)

Tip: Avoid the Snare body band (~200 – 500 Hz).

Q (1 — 8)

Bandpass sharpness.

- 1 – 1.5: broad, natural
- 1.5 – 3: standard
- 3 – 5: pinned
- 5 – 8: too sharp

Tip: Sweet spot 1.2 – 2.0.

DECAY (0.03 — 0.4 sec)

Length of the clap.

- 0.03 – 0.08: tight, tick
- 0.08 – 0.15: standard
- 0.15 – 0.25: long, rich
- 0.25 – 0.4: reverberant

Tip: STYLE × DECAY interact.

STYLE (0 — 1 (3 stages))

Burst pattern.

- 0.00 – 0.33: Single — classic
- 0.34 – 0.66: 707 — 2 bursts, 22 ms
- 0.67 – 1.00: 909 — 3 bursts, 13 ms

Tip: Slow tracks: Single. Techno: 707 / 909.

3. Character Presets

Goal	FREQ	Q	DECAY	STYLE
TR-808 (warm single)	1200	1.5	0.16	0.0 (Single)
TR-909 (bright triple)	1100	1.6	0.18	1.0 (909)
TR-707 (double)	1300	1.7	0.15	0.5 (707)
Deep House	1300	1.4	0.18	0.5 (707)
Acid (sharp)	1500	2.5	0.10	0.0 (Single)
Dub (long, spacious)	1100	1.5	0.30	0.0 (Single)
Modern Tech House	1400	1.8	0.12	0.5 (707)
Footwork	1400	2.0	0.10	0.0 (Single)
Industrial	1600	3.5	0.13	1.0 (909)

4. Tuning Order

#	Do this	Why
Step 1	Choose STYLE	Single or burst?
Step 2	Set FREQ for brightness	Where vs. the Snare?
Step 3	Set DECAY (length)	Tight or rich?
Step 4	Use Q to sharpen	Broad or pinned?
Step 5	PAN to place it	Offset to coexist with snare

5. Troubleshooting

Symptom	Cause & fix
Clashes with Snare	Pan by $\pm 0.2 - 0.3$, or separate FREQ by 200 – 300 Hz.
Doesn't sound like a clap	Keep Q in 1.2 – 1.8. Try STYLE 707/909.
Thin, weak presence	Switch to STYLE 909, or raise ClapVol.
Flammy bursts	Shorten DECAY (0.08 – 0.12), raise Q.
Boingy tone	Lower both FREQ and Q.

※ Clap and Snare often share the backbeat. Split by PAN or stack by FREQ for a thick layered backbeat.

Chapter 5

Bass

1. Internal Structure

Saw / Square oscillator + Sub oscillator -> Moog-style 4-pole Ladder Filter -> Drive -> LFO. TB-303 inspired, with sub-bass, chords, slide, accent and 4-voice polyphony.

Element	Role	Sonic character
Main oscillator	Saw (rich harmonics) or Square (hollow)	Fat vs. hard
Sub oscillator	Sine one octave below	Weight in the bottom
Ladder filter (Moog-style)	Lowpass 4-pole, capable of self-oscillation	Quack / squelch
Filter envelope	Sweeps the cutoff -- the "wow" of acid	TB-303 signature
Accent	Filter + amp boost on high-velocity notes	Accented notes pop
Glide	Portamento between notes (slide)	TB-303 "slide" feel
LFO	Cyclic modulation of cutoff or pitch	Wobble / vibrato
Bass Chord	Adds automatic harmony voices on top of one note	Maj / Min / Power / etc.

2. Parameter Reference

WAVE (Saw / Square)

Main waveform selection.

- Saw: rich harmonics, TB-303 signature
- Square: hollow / harder edge

Tip: Acid uses Saw. Square fits House pumping bass or 80s Synthwave.

CUTOFF (100 - 5000 Hz)

Ladder filter cutoff.

- 100-300 Hz: muffled, dub
- 300-800 Hz: standard round bass
- 800-2000 Hz: TB-303 sweet spot
- 2000-5000 Hz: filter nearly wide open

Tip: Interacts with EnvMod.

RESO (0 - 10)

Ladder filter resonance.

- 0-2: no resonance
- 2-5: natural push
- 5-8: clear "quack"
- 8-10: edge of self-oscillation

Tip: For extreme acid, push past 9.5.

ENV MOD (0 - 6000 Hz)

Filter envelope modulation depth.

- 0: static filter
- 500-2000: subtle "wow"
- 2000-4000: standard House
- 4000-6000: full sweep, acid

Tip: TB-303: CUTOFF 500-800, RESO 8-10, ENV MOD 4000-5500.

DECAY (0.05 - 1.5 sec)

Bass envelope decay.

- 0.05-0.15: acid pluck
- 0.15-0.30: standard acid
- 0.30-0.60: sustained House
- 0.60-1.5: dub bass, sub pads

Tip: Longer DECAY = slower filter motion.

ACCENT (0 - 1)

Accent boost on high-velocity notes.

- 0.0: accent disabled
- 0.3-0.5: subtle emphasis
- 0.5-0.8: clear accent
- 0.8-1.0: dramatic (Hardfloor)

GLIDE (0 - 0.2 sec)

Portamento between notes.

- 0.0-0.005: no glide
- 0.01-0.04: natural slide
- 0.04-0.10: gooey
- 0.10-0.20: dramatic SFX

Tip: Triggers only on notes marked "Slide".

DRIVE (0 - 1)

Saturation on output.

- 0.0: clean

- 0.1-0.3: subtle weight
- 0.3-0.6: acid signature
- 0.6-1.0: industrial bass

Tip: TB-303 sweet spot: 0.4-0.7.

SUB (0 - 1)

Sub oscillator mix.

- 0.0: no sub
- 0.2-0.4: subtle weight
- 0.4-0.7: solid sub
- 0.7-1.0: sub-dominated

Tip: Verify on full-range monitors / headphones.

LFO RATE / DEPTH / TGT (Rate 0.05-20 Hz, Depth 0-1)

LFO modulation.

- RATE 0.05-1 Hz: slow drift
- RATE 1-5 Hz: vibrato
- RATE 5-20 Hz: tremolo
- TGT = Cutoff: filter wobble
- TGT = Pitch: pitch vibrato (+/- 1 semitone)

Tip: Classic acid wobble: Rate 0.2-0.5, Depth 0.3-0.5, Cutoff.

CHORD (Off / +Oct / Power / Maj / Min / Sus4)

Automatic harmony.

- Off: mono (default)
- +Oct: adds an octave up
- Power: adds a perfect 5th
- Maj: major 3rd + 5th
- Min: minor 3rd + 5th
- Sus4: 4th + 5th

Tip: Chord ON = 3 voices per note. 4-voice polyphony budget.

3. Character Presets

Goal	WAVE	CUT	RESO	ENV	DEC	DRIVE	SUB
TB-303 acid (classic)	Saw	600	9.5	5500	0.28	0.55	0.0
Hardfloor (extreme acid)	Saw	480	9.5	5500	0.16	0.75	0.10
Deep House (fat, deep)	Saw	360	2.5	1200	0.55	0.10	0.50
Detroit Techno	Saw	500	3.5	2200	0.40	0.20	0.30
Dub bass (heavy, long)	Saw	300	2.0	1000	0.80	0.15	0.65
Sub bass (Trap, 808)	Saw	380	2.0	1500	0.70	0.20	0.70
House square	Sq	500	4.0	2500	0.40	0.30	0.30
Industrial (distorted)	Saw	550	6.0	3200	0.20	0.85	0.20
Synthwave (retro)	Saw	450	3.0	1500	0.50	0.20	0.40
Footwork (tight sub)	Saw	400	2.5	1400	0.60	0.18	0.60

4. Tuning Order

#	Do this	Why
Step 1	Pick WAVE (Saw or Square)	Saw = acid / lead; Square = funky / retro
Step 2	Shape with CUTOFF + RESO	Squelchy, or round?
Step 3	Set ENV MOD for motion	Static or dynamic filter?
Step 4	DECAY for length	Fit it to the space in the track
Step 5	DRIVE for forwardness	Clean or grit
Step 6	SUB for low-end weight	Survives small speakers
Step 7	LFO / CHORD for flavour	Wobble and harmony

5. Troubleshooting

Symptom	Cause & fix
Bass feels "not there"	CUTOFF too low, SUB too high. Lower SUB, raise CUTOFF.
Kick and bass mud together	Raise SC AMT to 0.4-0.5, or separate CUTOFF from kick PITCH.
No acid "squelch"	Try RESO 8.5+, ENV MOD 4000+, DECAY 0.20-0.30.
Slide does not work	GLIDE too short. Start 0.03-0.06.
Chord sounds thin	Chord ON uses more voices. 4-voice limit.
Higher DRIVE = quieter	Tanh normalises level. Compensate with Bass Vol.
LFO inaudible	Check that LFO DEPTH is not 0.

※ The bass-kick interaction is the most important in the mix. Always design the bass against the kick. Separate CUTOFF from the kick PITCH, and use SC AMT to duck the bass under the kick - these two tricks resolve most low-end issues.